Vivan

Use Case 01: Award XP on Task Completion

Primary Actor: User

Stakeholders and Interests:

- User: wants XP to feel “right”
- Product Team: wants XP rules that drive engagement
- System Admin: wants XP awards logged

Preconditions:

- User is authenticated
- Task exists and not completed
- XP rule configuration is active

Success guarantee (Postconditions):

- XP is added exactly once
- XP event is recorded
- Updated XP is visible in UI

Main success scenario:

1. User marks task as completed.
2. System verifies task eligibility.
3. System calculates XP.
4. System applies eligible multipliers if needed.
5. System records XP event.

6. System updates user's total XP.
7. System updates UI in real time.

Extensions:

- 2a. Task already completed, System denies XP and shows message.
- 3a. XP rule missing, Default XP applied and error logged.
- 5a. Database failure, XP not awarded user notified

Special requirements:

- XP award must be one time thing

Technology and data variation list:

- Web client
- Mobile client
- Offline sync mode

Use Case 02: Level Up via Progression Thresholds

Primary Actor: User

Stakeholders and Interests:

- User: wants satisfying progression pacing
- Product Team: wants adjustable threshold system

Preconditions:

- User has XP value stored
- Level thresholds configured

Success guarantee:

- User level reflects correct XP range
- Level-up event recorded
- UI reflects new level immediately

Main success scenario:

1. XP total changes.
2. System compares XP to threshold table/formula.
3. If threshold crossed, system updates level.
4. Level-up event recorded.
5. UI displays updated level and progress bar.

Extensions:

- 3a. Multiple levels crossed, All applied in single update.
- 2a. Threshold configuration updated, Level recalculated.

Special requirements:

- Level must be deterministically derived from XP.

Technology and data variation list:

- Threshold table stored in database
- Formula-based progression system

Use Case UC 03: Track Daily and Weekly Streaks

Primary Actor: User

Stakeholders and Interests:

- User: wants streak rewards without harsh punishment

- Product Team: wants configurable grace rule

Preconditions:

- User authenticated
- Time zone configured
- At least one eligible task completed

Success guarantee:

- Daily streak updated correctly
- Weekly streak updated correctly
- Streak state visible in UI

Main success scenario:

1. User completes an eligible task.
2. System checks if streak already counted today.
3. System updates daily streak.
4. System updates weekly streak.
5. System applies grace rule if applicable.
6. UI updates streak display.

Extensions:

- 2a. Multiple completions same day, Only first counts.
- 5a. Grace rule triggered, Streak preserved.

Special requirements:

- Streak calculation must be consistent across devices.

Technology and data variation list:

- Server-side time validation
- Client time display

Use Case UC 04: Apply Bonus Multipliers

Primary Actor: User

Stakeholders and Interests:

- User: wants noticeable but fair bonuses
- Product Team: wants capped multiplier system

Preconditions:

- Task completed
- Base XP calculated
- Multiplier rules configured

Success guarantee:

- Final XP calculated correctly
- Multiplier breakdown visible
- XP event logged

Main success scenario:

1. System calculates base XP.
2. System checks streak multiplier eligibility.
3. System checks early completion bonus eligibility.
4. System combines multipliers

5. Final XP awarded.
6. UI displays XP breakdown.

Extensions:

- 4a. Multiplier exceeds cap, Cap applied.
- 3a. Missing due date, Early bonus skipped.

Special requirements:

- A multiplier system must prevent abuse/exploit attempts.

Technology and data variation list:

- Configurable multiplier rules in database

Supplementary Specification

NFR 01: Real Time XP and Progress Updates

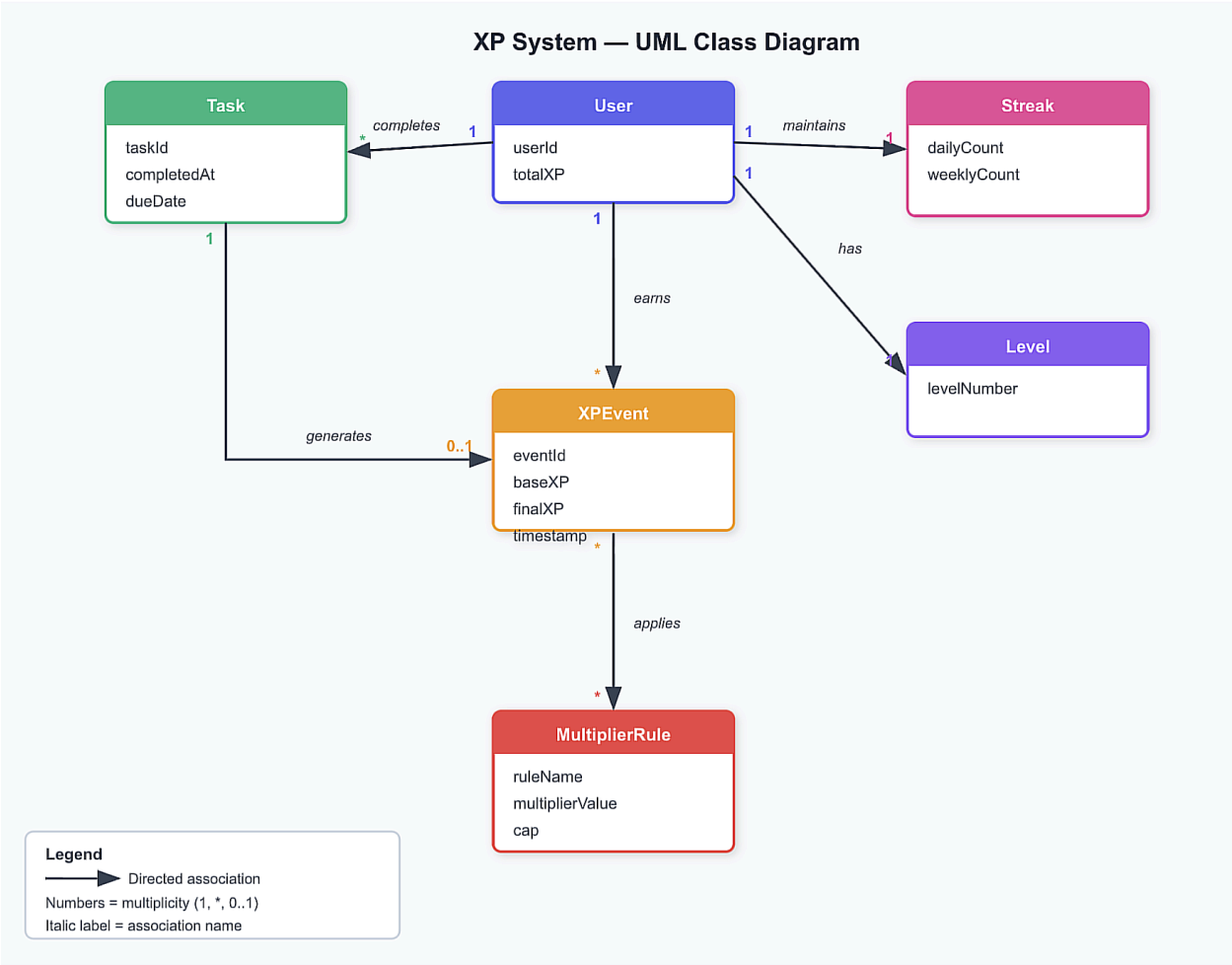
Category: Performance / User Experience

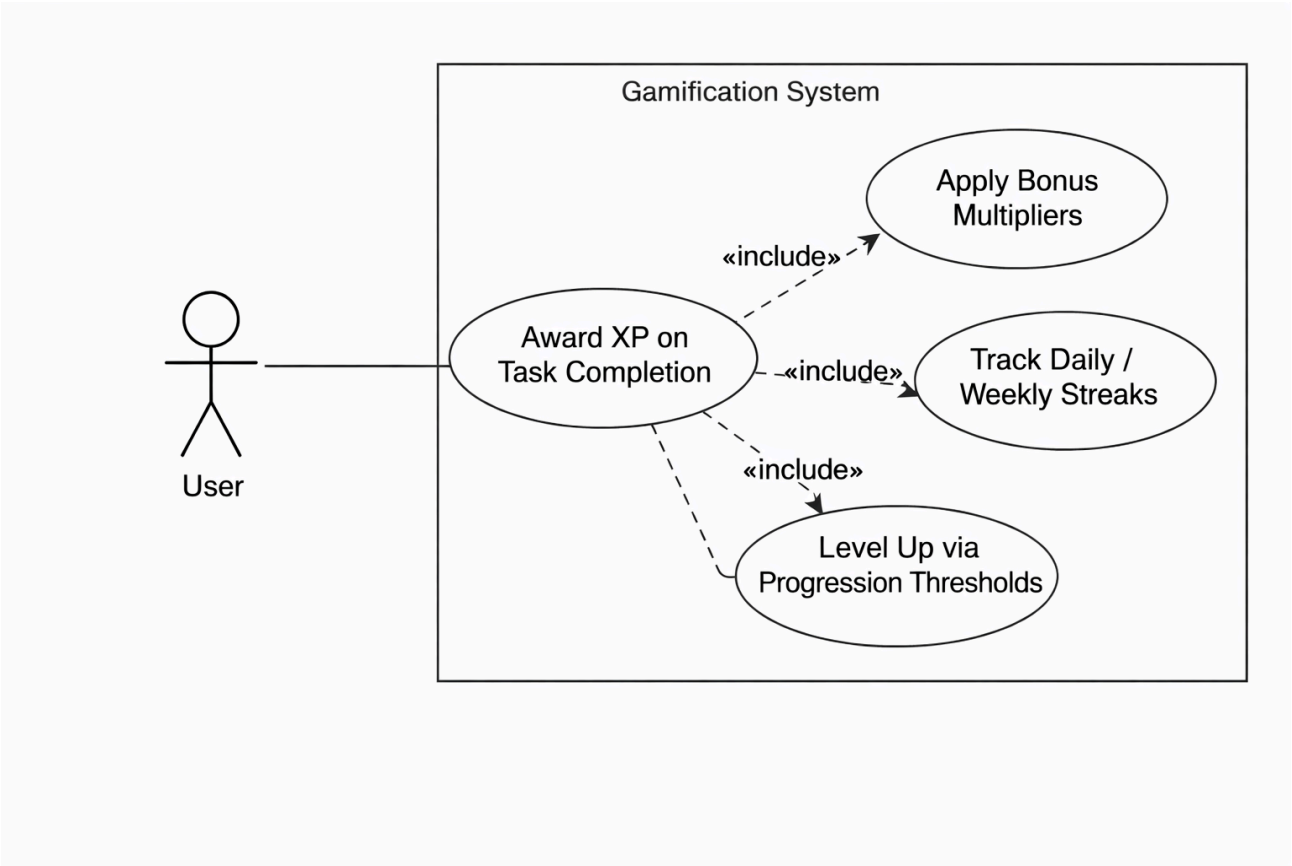
Requirement:
When a user's XP, level, or streak changes, the updated values must appear consistently across all UI surfaces immediately.

Explanation:
Progression feedback must feel instant to maintain engagement.

Acceptance Criteria:

- XP/level updates visible within 1 second.
- No inconsistent values between pages.
- Offline mode shows syncing state until confirmed.


XP System — UML Class Diagram
Task
taskId
completedAt
dueDate
User
userId
totalXP
Streak
dailyCount
weeklyCount
Level
levelNumber
XPEvent
eventId
baseXP
finalXP
timestamp
MultiplierRule
ruleName
multiplierValue
cap
completes
maintains
has
earns
generates
applies
1
*
0..1
Legend
Directed association
Numbers = multiplicity (1, *, 0..1)
Italic label = association name


Gamification System
Apply Bonus Multipliers
«include»
Award XP on Task Completion
«include»
Track Daily / Weekly Streaks
«include»
Level Up via Progression Thresholds
User